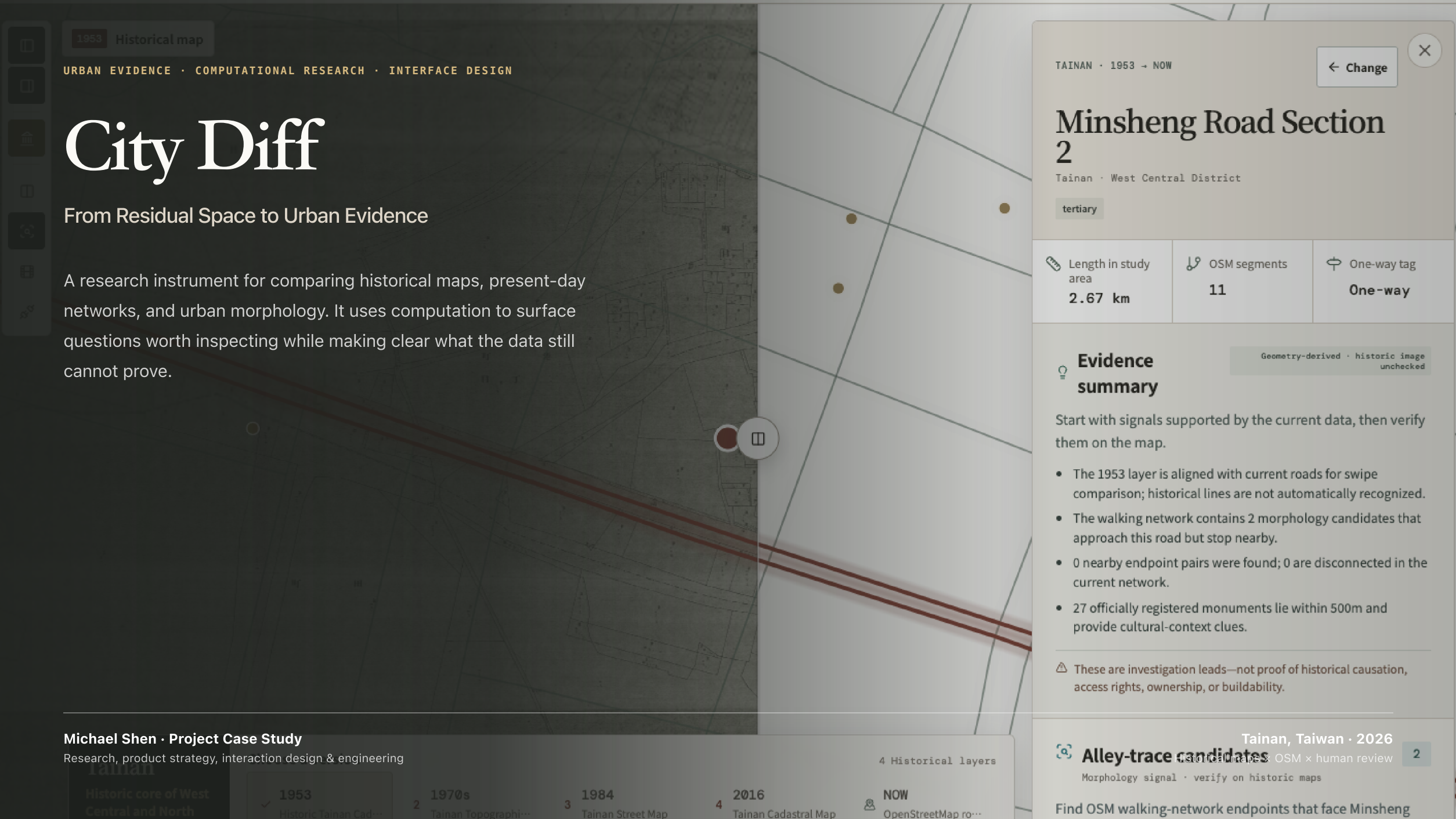




1953 Historical map

URBAN EVIDENCE · COMPUTATIONAL RESEARCH · INTERFACE DESIGN

City Diff

From Residual Space to Urban Evidence

A research instrument for comparing historical maps, present-day networks, and urban morphology. It uses computation to surface questions worth inspecting while making clear what the data still cannot prove.

Michael Shen · Project Case Study
Research, product strategy, interaction design & engineering

4 Historical layers

1

Tainan

Historic core of West Central and North

2

1953

Historic Tainan Cad...

3

1970s

Tainan Topographi...

4

1984

Tainan Street Map

5

2016

Tainan Cadastral Map

6

NOW

OpenStreetMap ro...

TAINAN · 1953 → NOW

← Change

×

Minsheng Road Section 2

Tainan · West Central District

tertiary

📏 Length in study area

2.67 km

📍 OSM segments

11

🚦 One-way tag

One-way

💡 Evidence summary

Geometry-derived · historic image unchecked

Start with signals supported by the current data, then verify them on the map.

- The 1953 layer is aligned with current roads for swipe comparison; historical lines are not automatically recognized.
- The walking network contains 2 morphology candidates that approach this road but stop nearby.
- 0 nearby endpoint pairs were found; 0 are disconnected in the current network.
- 27 officially registered monuments lie within 500m and provide cultural-context clues.

⚠️

These are investigation leads—not proof of historical causation, access rights, ownership, or buildability.

📍 Tainan, Taiwan · 2026

Alley-trace candidates

Historic maps × OSM × human review

2

Morphology signal · verify on historic maps

Find OSM walking-network endpoints that face Minsheng

From proposing interventions to examining how “broken” is defined

City Diff returns to a 2021 architecture thesis on residual spaces in Tainan and reframes its design premise as a testable, contestable research question.

2021

Propose repairs first

The Imperfect City read odd lots and residual spaces as products of overlapping urban systems, then proposed distributed micro-interventions.

Architecture thesis

Spatial intervention



2026

Test the evidence first

Translate morphological intuition into an algorithm, then use historical maps, source records, and human judgment to inspect each candidate site.

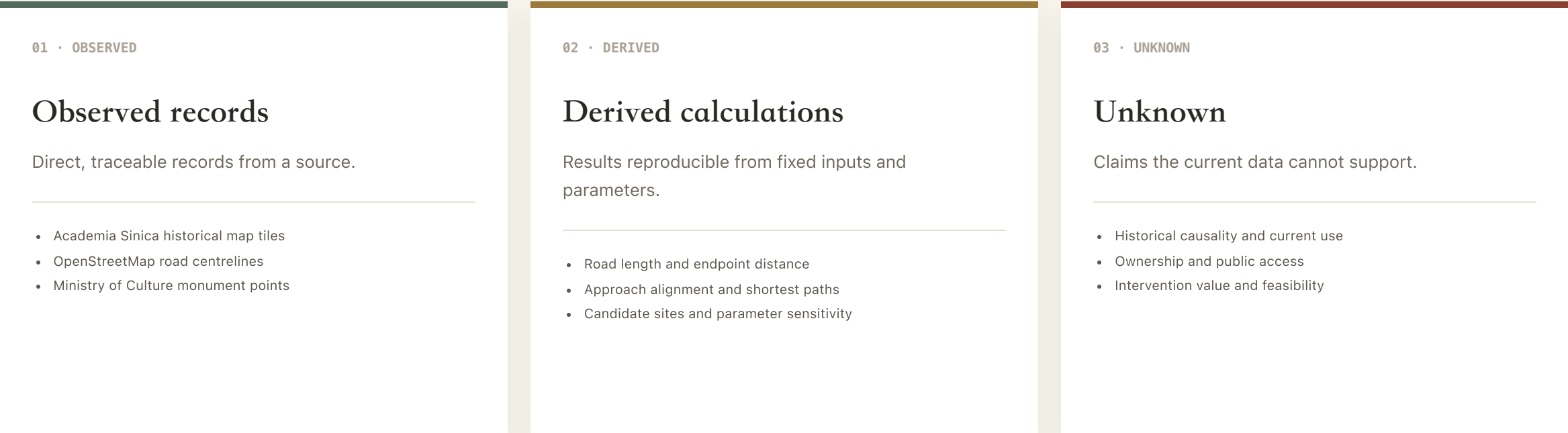
Computational research

Human in the loop

Can a human-in-the-loop spatial tool produce inspectable hypotheses about urban discontinuity without mistaking computational proxies for urban facts?

The interface separates records, calculations, and claims into three layers

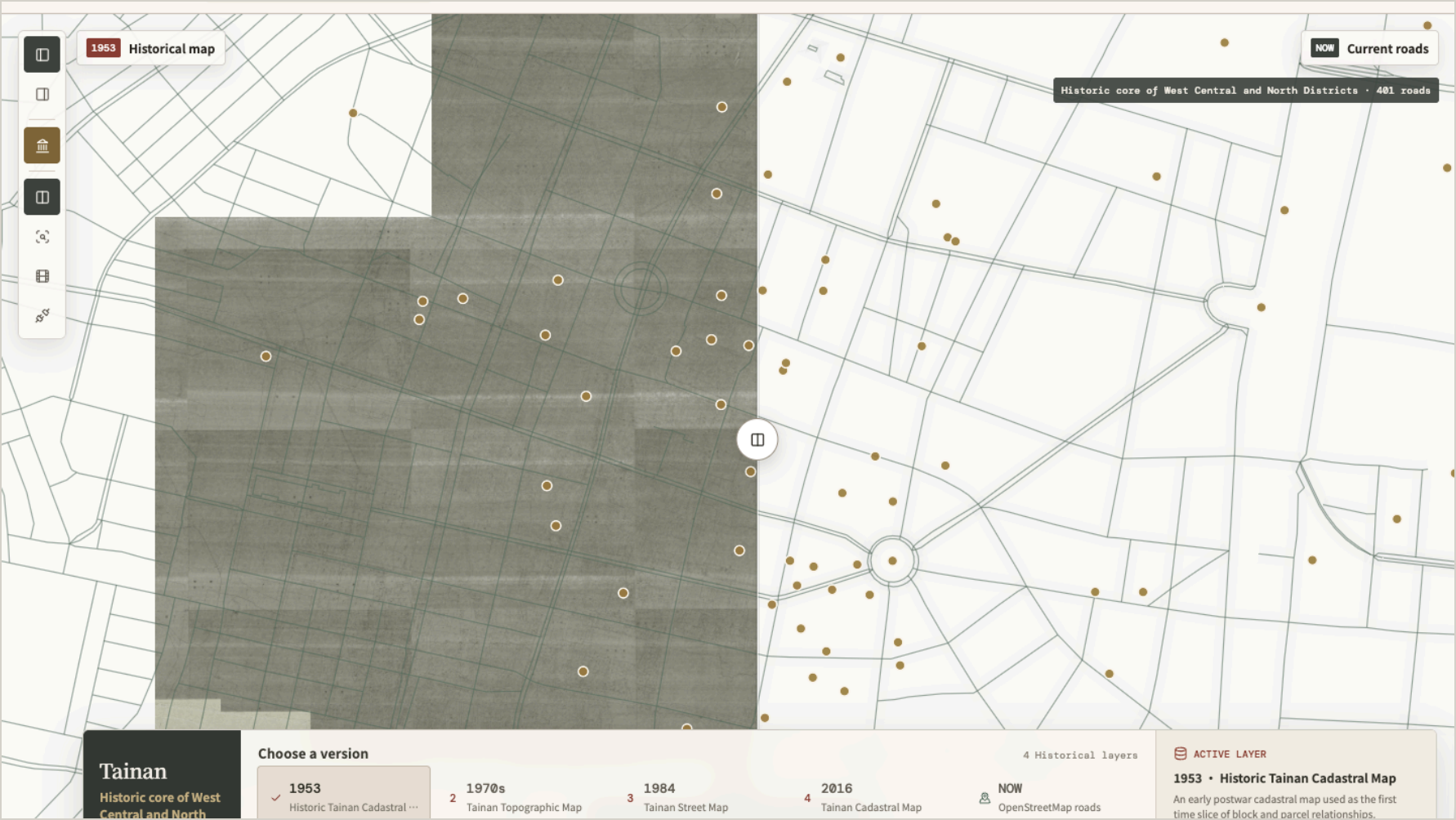
This is the product’s ethical framework: users must be able to see what the system knows, how it derived a result, and what remains missing.



- 4 historical city packs
- 9 explorer cities
- Source timestamps
- SHA-256 provenance
- Explicit limitations

One coordinate, four historical time slices, and the present-day network

A draggable divider preserves spatial orientation while the timeline keeps each map's date, type, and source in one workspace.



TAINAN · 1953 CADASTRAL MAP ↔ PRESENT-DAY OPENSTREETMAP

- 1953 · Cadastral map**

An early postwar view of block and parcel relationships.
- 1970s · Topographic map**

Shows roads, blocks, and built fabric without treating them as legal boundaries.
- 1984 · Street map**

A comparison point after further urban road expansion.
- 2016 · Cadastral imagery**

A newer comparison layer that is still not live legal data.

Enter a traceable research workflow through one street

After a user selects a real street, the interface presents a geometric summary and brings historical imagery, cultural context, and candidate sites into one review workflow.



2.67

km study-area
length

11

OSM
segments

27

heritage
records within
500 m

Read the evidence summary

Place supported signals and unsupported conclusions in the same view.

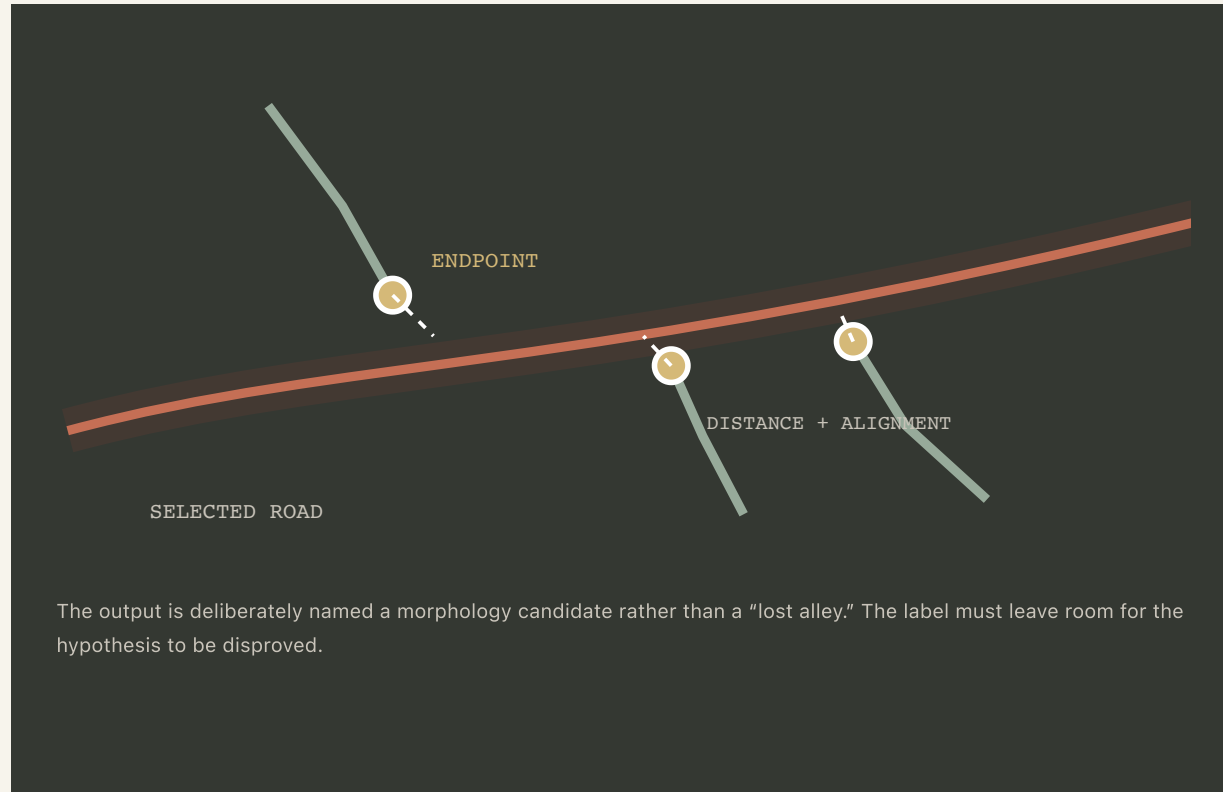
Return to the map

The highlighted corridor guides visual inspection without claiming a legal boundary or historical cause.

Complete a human review

Save the candidate, data snapshot, historical layer, and observation record together.

Turn morphological intuition into geometric rules that can be challenged



The algorithm does not judge the city. It narrows 4,800 walking-network segments into sites worth checking against historical maps.

01

Build the network graph

Merge nearby nodes using an 8 m tolerance.

02

Find endpoints

Keep degree-one nodes away from the study boundary.

03

Measure distance

Require endpoints to be 3–55 m from the selected road.

04

Compare direction

Require cosine alignment toward the road ≥ 0.35 .

05

Remove duplicates

Within 12 m, retain only the stronger alignment score.

06

Send to human review

Check whether a connection is visible in historical and current records.

4,800 network segments

→ 379

unique endpoint sites

Scan the full study area instead of selecting only compelling cases

The same rules evaluate all 401 named roads in the Tainan study area, retaining zero-result roads as controls.

Tainan roads ranked by computational signals

Counts are morphology hypotheses, not verified historical alleys or intervention priorities.



401

named roads fully analyzed

193

roads with at least one morphology or connectivity signal

379

unique endpoint sites

30

urban stitch candidate pairs

Count is not value.

The ranking shows where the algorithm raises more questions. It does not measure public value, historical importance, or investment priority.

Zero is also a result.

Chenggong Road has no candidates under the baseline parameters. This does not mean it has no urban impact; the current proxy simply did not detect the specified morphology.

Parameters are spatial positions taken by the researcher

Forty-eight parameter combinations test node tolerance, search distance, and alignment thresholds, exposing how unstable the output can be.

27
–270

A tenfold candidate range across the same roads

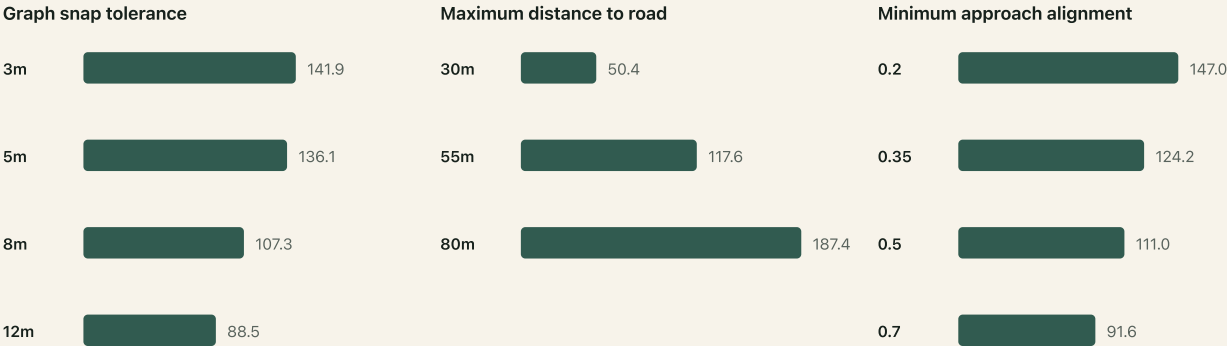
Search distance has the largest effect. Fifty-five metres is not objective truth; it defines the scale at which the system interprets a site as related to a road.

No ground truth → robustness, not accuracy.

This is not an accuracy test. It measures sensitivity to research choices and requires the interface to disclose them.

Candidate counts are highly parameter-sensitive

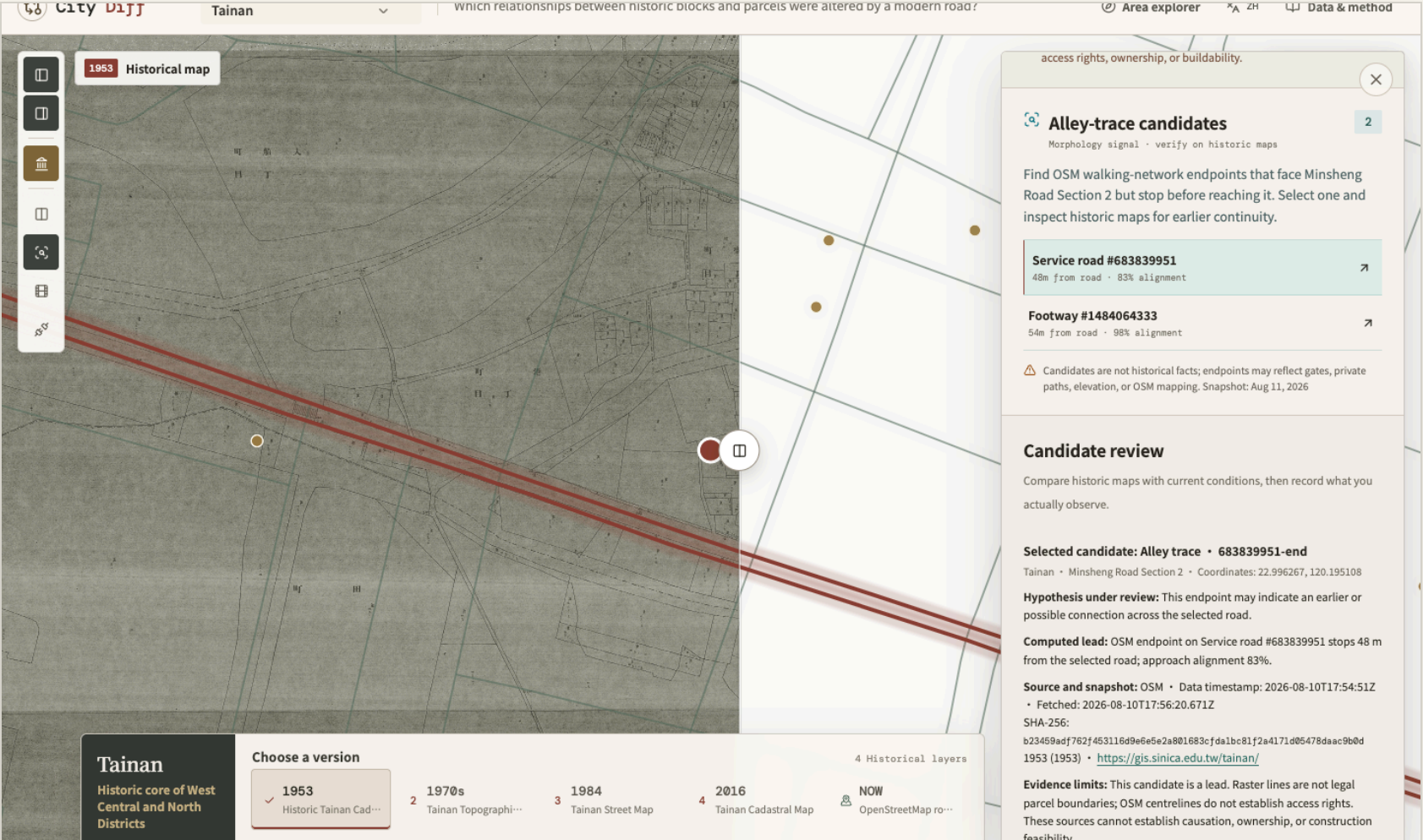
Mean signals across 30 longest named roads; all 48 combinations range from 27 to 270 candidates.



Lower snap tolerance, larger search distance, and looser alignment generally surface more hypotheses. This is a robustness warning, not an accuracy score.

The algorithm narrows the search; people interpret and leave an accountable record

Each candidate retains its data snapshot, parameter basis, historical layer, human observations, and specific unresolved questions.

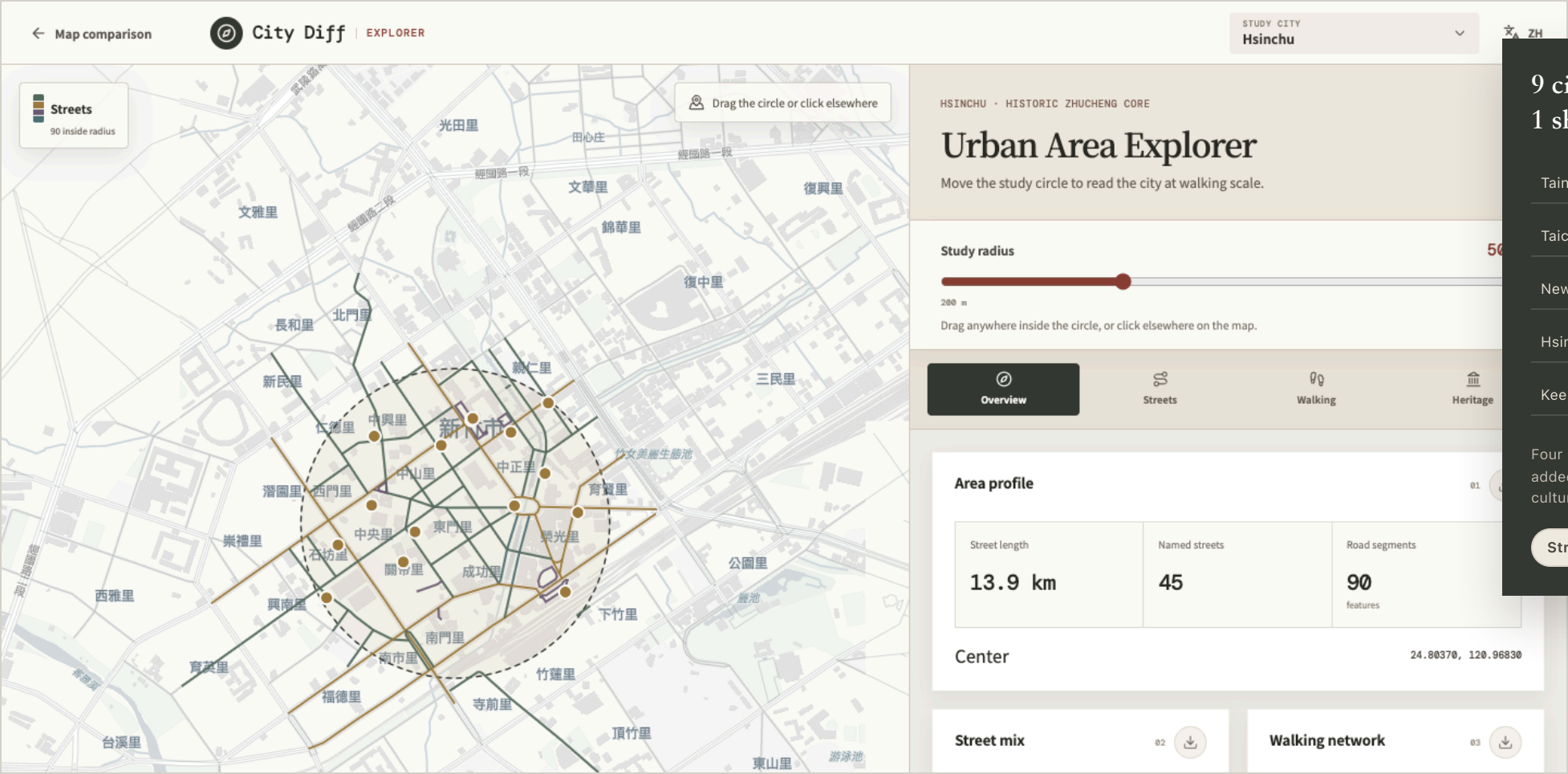


CANDIDATE · OSM ENDPOINT → HISTORICAL IMAGE → HUMAN REVIEW

- 1 Locate the candidate**
Record coordinates, distance, and alignment score.
- 2 Cross-check evidence**
Inspect historical maps and the current network at the same coordinates.
- 3 Record observations**
Separate what is visible, cited sources, and unresolved questions.
- 4 Preserve the snapshot**
Use timestamps and SHA-256 to identify changed source data.
- 5 Export an auditable case**
Export JSON or Markdown so others can inspect the reasoning.

Extend the method into a walking-scale reader for nine cities

Move the study circle to recalculate street hierarchy, walking networks, surface tags, and heritage context without selecting a road first.



9 cities
1 shared method

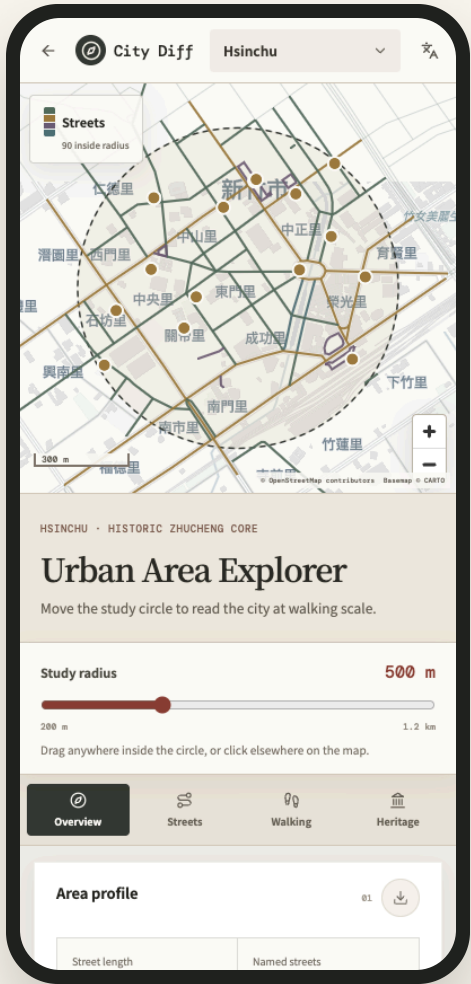
Tainan	Kaohsiung
Taichung	Taipei
New Taipei	Taoyuan
Hsinchu	Chiayi
Keelung	

Four cities include historical layers; five added cities use the same OSM and cultural-heritage analysis method.

Street mix Walking

One research tool for desktop inspection and mobile exploration

Mobile retains direct circle dragging, clear radius controls, and single-column cards; desktop keeps the map and analysis visible together.



MOBILE UX

Reduce the viewport, not the information needed for judgment

The study area, data lenses, and key values remain near the first screen. Touch targets grow, cards become a single column, and the circle stays directly draggable.

30% mobile sheet

Preserves map context after road selection and expands upward on demand.

6 profile cards

Area profile, street mix, walking network, hierarchy, surfaces, and heritage.

Real-time recompute

All metrics update when the city, center, or radius changes.

Every card exports as a 2× PNG

city · study area · radius · current values

Detected is not the same as proven.

Urban computation should reveal what still needs to be negotiated.

WHAT I BUILT

An operational urban evidence interface

Integrates historical maps, present-day networks, cultural heritage, candidate algorithms, human review, and a multi-city Explorer.

WHAT I LEARNED

Uncertainty must be designed

Parameters, data gaps, and zero results must remain visible; otherwise a polished interface only amplifies false certainty.

WHAT COMES NEXT

From geometric candidates to fieldwork and governance

The next phase requires vector cadastral data, field observation, local knowledge, access rights, and governance processes.